

The NEED FOR CALIBRATION And How To Improve It



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Mistakes are an integral part of human life. Some mistakes can have a larger impact than the others, and this calls for an application to be tested, inspected and certified before handing it over for usage. This has led to a lot of standards and regulations to come up. Electrotechnical Assessment Scheme (EAS), for example, governs the requirements of a suitable system to ensure consistency of instruments.

Since most test instruments involve electronics, a lot can go wrong due to the sensitive components involved. This requires regular calibration of equipment to keep these up to date. However, the bigger issue is the compromise of performance in handheld equipment over benchtop ones.

Handheld vs benchtop equipment

"There are certain challenges in reducing the form factor while providing better performance and specifications," says V. Mahadevan, managing director, VigVen Tech

Points to remember

Some factors that can help in improving measurements are:

- Ensure calibration of every measuring instrument used for testing, including the ones used at the design stage.
- Concurrent approach to engineering and design for testability must be followed. This means using the right tools without any handicap.
- Use of advanced simulation tools and ensuring integration of board design with EMI/EMC as well as mechanical CADD packages.
- Quality over compromise should be the order of the day with each individual onboard the design bus.

Mark Pvt Ltd. Reducing the form factor means removing certain components and compromising in certain areas.

Handheld equipment minimise board design and, hence, avoid components that generate too much heat. This allows removal of heat-sinks and helps the



*Proper calibration
requires fine-tuning
the measurements*